

Afnan Sherif . Jarvis Consulting

Software Developer with 5+ years of experience building backend applications, REST APIs, and AI/ML solutions using Python, SQL, PostgreSQL, Docker, and Linux. Experienced in developing and supporting production software, optimizing database performance, troubleshooting backend issues, and integrating machine learning into practical applications. Developed an AI-based phishing and malware detection system, combining machine learning, API development, and cybersecurity to address a real-world problem. Currently expanding my knowledge of cloud security and AI engineering through hands-on projects and AWS Cloud Security Foundations training. I enjoy learning new technologies, solving complex problems, and building reliable, secure software that creates practical value.

Skills

Proficient: Python, SQL/PostgreSQL, Agile/Scrum, AI/ML, REST API, Git/GitHub, C#/.NET

Competent: Docker, Linux/Bash, Database Management, AWS/GCP, Flask, CI/CD, Testing & Debugging

Familiar: Java, HTML/CSS, Cloud Computing, Django, Flask, Monitoring and Logging

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_AfnanSherif

Linux Cluster Monitoring Agent [GitHub]: This project implements an automated Linux monitoring solution using Bash scripting to collect host-level metrics such as CPU utilization and memory usage. A lightweight monitoring agent runs using Crontab on each host at regular intervals to capture real-time performance data and store it in a centralized PostgreSQL database running inside a Docker container, along with logs for troubleshooting and tracking.

Core Java - Grep App [GitHub]: This project is a Java application that mimics the Linux grep command by recursively searching files and extracting lines that match a given regular expression. Two implementations were developed: JavaGrepImp using core Java and JavaGrepLambdaImp using Java Streams and lambda expressions. The project utilizes core Java concepts such as OOP, collections, file I/O, exception handling, recursion, and regular expressions. Additional tools and technologies used include Maven for dependency management, SLF4J/Log4j for logging, IntelliJ IDEA as the IDE, Git/GitHub for source control, and Docker for application distribution.

Python Data Analytics [GitHub]: This project focused on analyzing historical retail transaction data for a UK-based e-commerce company to identify customer behavior patterns and support business growth decisions. The goal was to improve customer retention, revenue growth, and marketing strategy through data-driven insights. The project was implemented in Python using a Jupyter Notebook environment. Pandas was used for data cleaning and transformation, NumPy for statistical analysis, and Matplotlib for visualization. The analysis covered sales trends, customer activity, revenue patterns, and customer segmentation to identify high-value customers and purchasing behavior insights.

Highlighted Projects

IoT Monitoring & Threat Detection | Raspberry Pi, AWS IoT, EC2, Suricata, ELK [GitHub]: Built an end-to-end IoT monitoring system. Used a Raspberry Pi with an MCP9808 temperature sensor and connected it securely to AWS IoT Core using device certificates and MQTT over TLS. I also deployed an EC2 instance running Mosquitto, Suricata, Filebeat, Elasticsearch and Kibana for security monitoring. VPC Flow Logs were stored in S3, which allowed me to investigate network traffic and identify high-volume MQTT activity. We also had to troubleshoot resource limitations on the EC2 instance and tune the Elasticsearch heap because I was running several services on a relatively small instance.

Detecting Malware using Machine Learning [GitHub]: The project focused on developing an intelligent email threat detection system designed to identify phishing and malicious emails by integrating scikit-learn machine learning models. It involved collecting and preparing datasets, training and evaluating machine learning models using metrics such as accuracy, precision, recall, and F1-score, and continuously improving the models to reduce false positives. I also developed a Python-based web application, REST API, and browser extension to enable real-time email scanning and threat detection directly within the user interface. This project was especially meaningful because it combined my interests in cybersecurity, software development, and artificial intelligence to solve a real-world problem.

Stock Quote App [GitHub]: The project focused on developing a Java-based command-line stock trading application that retrieves stock information from the Alpha Vantage API and stores quote and portfolio data in PostgreSQL using JDBC. I designed the application with a layered structure consisting of controllers, services, DAOs, HTTP helpers, DTOs, and a database connection manager. The application allows users to view stock quotes, buy and sell shares, and manage

their portfolio. I used Jackson for JSON-to-Java mapping, OkHttp for API requests, JUnit and Mockito for testing, and SLF4J/Log4j for logging. I also packaged the application with Maven and containerized it using Docker, running the Java application with Eclipse Temurin JDK 17 and PostgreSQL in a separate container connected through a custom Docker network. The final Docker image was published to Docker Hub. This project gave me hands-on experience building and connecting the different components of a complete Java application, from external API communication and database persistence to testing and containerization.

Professional Experiences

Software Developer, Jarvis (03/2026-present): Apply my technical and professional skills through hands-on training, real-world projects, and collaboration in a fast-paced consulting environment. Use technologies such as PostgreSQL, Linux, CLI tools, and Docker while building software solutions, managing databases, solving problems, and completing practical assignments that reflect client needs and industry standards.

Sales Representative, Kognitive Sales Solutions (04/2025-03/2026): Promoted credit card products in high-traffic retail environments by engaging customers, explaining product benefits, and building strong client relationships. Consistently used communication, persuasion, and problem-solving skills to meet sales targets, handle objections, and deliver a positive customer experience in a fast-paced setting.

Software Engineer (Part-time), Sentinel360 (08/2024-02/2026): Developed and maintained AI-driven phishing detection systems in a production environment that scanned hundreds of websites daily for suspicious behavior, malicious patterns, and phishing indicators. Improved application performance by optimizing database queries, refining caching logic, troubleshooting backend issues, and enhancing existing Flask/Django features for better scalability and reliability.

Full Stack Developer, NBS Venture (12/2022-01/2024): Developed full-stack web applications for the restaurant sector using HTML, CSS, and C#, improved internal restaurant management tools to meet operational requirements, and automated testing and deployment workflows to increase efficiency and reliability.

Junior Software Developer, Arib (01/2021-11/2022): Assisted in developing and maintaining backend platform services for business management applications within an Agile environment by supporting feature releases, troubleshooting application issues, and improving system functionality. Created automated Python testing scripts to identify defects earlier in development, reduce recurring bugs, and developed data-processing scripts to clean datasets and improve data accuracy.

Backend Developer (Part-time), Kolay Pro (01/2020-12/2020): Developed RESTful APIs and backend solutions for digital business applications using Python and Flask, optimized PostgreSQL queries to improve response times, and collaborated with cross-functional teams to deliver efficient backend services.

Education

Seneca Polytechnic College (2025-2026), Cybersecurity and Threat Management Certification, Computer Science - GPA: 3.7/4.0

Istanbul Kultur University (2019-2022), Computer Engineering, Electrical and Computer Engineering - GPA: 3.3/4.0

Miscellaneous

- AWS Cloud Security Foundations - Trained
- Microsoft Azure AI Fundamentals (2025)
- Certificate of Participation - Human-Computer Interaction, Optimization, and Robotic Applications (HORA) 2022 (supported by IEEE). Presented and published paper Detecting Phishing Websites using Machine Learning: <https://ieeexplore.ieee.org/abstract/document/9799917>
- I use photography to capture meaningful moments and savor them forever.
- I love travelling because I get to explore new places, meet new people, experience different cultures, and enjoy beautiful scenery.
- I like to learn a new skill every 3-6 months.
- Some of my hobbies are crocheting, gardening, cooking Italian cuisine, and DIY crafts.
- I am allergic to cats, but I love them, especially my cat Nemo.